Plasmid Extraction

M07 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Purify plasmid DNA from 5-15 mL of overnight bacterial culture using the Vazyme FastPure EndoFree Plasmid Mini Plus Kit (DC204). The workflow uses alkaline lysis, silica-membrane binding and impurity removal to prepare DNA for routine molecular biology and cell transfection. [1]

Materials & Equipment

Item	Manufacturer / Cat. no.	Use / Specification
FastPure EndoFree Plasmid Mini Plus Kit	Vazyme / DC204-01	RNase A Solution; Buffers QB, P1, P2, N3, PW1 and PW2; Endotoxin-free Elution Buffer; FastPure DNA Mini Columns III; 2 mL Collection Tubes
Overnight bacterial culture	Plasmid-bearing E. coli	5-15 mL per standard preparation
Absolute ethanol	Sangon Biotech	Prepare PW2 using the bottle-label volume
Isopropanol	Molecular biology grade	DNA binding
Sterile centrifuge tubes; pipettes and tips	Axygen tips; laboratory tubes and pipettes	1.5, 2 and 5 mL tubes; suitable harvest vessels
Centrifuges and compatible rotors	Eppendorf centrifuge; laboratory equipment	13,400 × g for the standard workflow; 4,000 × g for the alternative harvest
Vortex mixer; heating block or water bath	Kylin-Bell mixer; laboratory heating equipment	Pellet resuspension; 37°C for buffer preparation; optional 65°C elution-buffer preheating

Before You Start

Run the extraction at room temperature. Use 5-15 mL of overnight culture for the standard procedure; the manufacturer recommends 12-14 h of growth and advises against exceeding 16 h. [1]

Before first use, add the entire supplied RNase A Solution to Buffer P1 and mix. Store the supplemented P1 at 2-8°C for up to 6 months. Prepare Buffer PW2 with the volume of absolute ethanol stated on its bottle label. [1]

Check Buffers QB and P2 for crystals. Dissolve any precipitate at 37°C, then cool to room temperature and mix. When room temperature is 20°C or below, warm P2 at 37°C for 10 min until fully dissolved before cooling and use. [1]

Procedure

Standard 5-15 mL workflow, based on the DC204 V24.2 manual, section 08-2. [1]

A. Harvest, lyse and clarify

1. Pellet 5-15 mL of overnight bacterial culture at 13,400 × g for 1 min. Alternatively, collect the cells at 4,000 × g for 15 min. Remove as much residual culture supernatant as possible.

2. Add 500 µL Buffer P1 containing RNase A. Vortex until the pellet is fully dispersed.

3. Immediately add 500 µL Buffer P2. Mix with 8-10 gentle inversions and leave at room temperature for 3 min. The suspension should become clear, viscous and blue. Do not vortex, and keep the total lysis time within 5 min.

4. Add 500 µL Buffer N3 and immediately invert gently 12-15 times, until all blue color disappears and a white flocculent precipitate forms. Leave at room temperature for 10 min, then centrifuge at 13,400 × g for 10 min.

5. Transfer the clarified supernatant (approximately 1.5 mL) to a 5 mL tube. Add isopropanol equal to 0.3 times the actual recovered supernatant volume (450 µL for 1.5 mL), and mix with 10-15 inversions.

B. Equilibrate and load the column

6. Place a FastPure DNA Mini Column III in a 2 mL Collection Tube. Apply 200 µL Buffer QB, centrifuge at 13,400 × g for 1 min and discard the flow-through. Use the equilibrated column the same day.

7. Load up to 750 µL of the mixture from step 5 onto the column. Centrifuge at 13,400 × g for 1 min and discard the flow-through.

8. Repeat step 7 with the remaining mixture until the entire sample has passed through the same column.

C. Wash, dry and elute

9. Apply 500 µL Buffer PW1. Centrifuge at 13,400 × g for 1 min and discard the flow-through.

10. Apply 600 µL Buffer PW2 prepared with ethanol. Centrifuge at 13,400 × g for 1 min and discard the flow-through.

11. Repeat step 10 once, giving two PW2 washes in total.

12. Return the column to the collection tube and centrifuge the empty column at 13,400 × g for 2 min to dry the membrane.

13. Move the column to a clean 1.5 mL tube. Dispense 60-200 µL Endotoxin-free Elution Buffer onto the center of the membrane. Wait 1 min at room temperature, then centrifuge at 13,400 × g for 1 min. Keep the eluate and discard the column.

14. Store the extracted plasmid DNA at -30 to -15 °C.

p.s.

Mix immediately but gently after adding P2 or N3. Persistent turbidity during lysis can indicate excessive biomass; do not compensate by extending lysis beyond 5 min. The clarified supernatant may contain a few tiny floating white particles, which the manual states do not prevent further processing. [1]

Calculate isopropanol from the recovered supernatant volume rather than automatically adding 450 µL; excess isopropanol can increase RNA carryover. [1]

Do not reduce the elution volume below 60 µL. The manual allows preheating elution buffer to 65°C or reapplying the collected eluate to improve recovery. If using ddH2O instead, its pH must be 7.0-8.5. [1]

Expected Outcome

Collect plasmid DNA in the final eluate. During extraction, the visible checkpoints are a clear blue lysate after P2 and disappearance of blue color with white precipitate after N3. Yield varies with plasmid copy number, host strain and culture conditions. [1]

Troubleshooting

Problem	Check first	Suggested action
Low DNA recovery	Plasmid copy number/size, culture age, P2 precipitation and column equilibration.	Dissolve buffer precipitates and perform the QB step. For a low-copy plasmid, the manual permits 30 mL culture with proportionally increased P1, P2 and N3. Preheat elution buffer to 65°C for low-copy or >10 kb plasmids. [1]
Genomic DNA contamination	Mixing intensity after P2 and total lysis duration.	Use gentle inversion after P2; keep lysis within 5 min, including when processing multiple samples. [1]
Low purity / suspected salt or ethanol carryover	Wash coverage and completion of the dry spin.	Distribute wash buffer around the inner column wall. If ethanol carryover is suspected, leave the column open at room temperature for 5 min after the dry spin, before elution. [1]
RNA contamination	RNase A addition/storage, biomass and isopropanol volume.	Use correctly supplemented P1 stored at 2-8°C; reduce excess biomass. Add only 0.3 volumes of isopropanol, based on the actual supernatant volume. [1]

Reference

[1] Vazyme Biotech Co., Ltd. FastPure EndoFree Plasmid Mini Plus Kit, DC204, Instruction Manual, Version 24.2 (Chinese). Sections 02-06 and 08-09; PDF pp. 3-6. Accessed 2026-09-19. Open source